

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$4x + 12 = \frac{a(x+3)}{2}$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

Answer

- A. 0
- B. 3
- C. 8
- D. 12

Correct Answer: C

Rationale

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = a(x + 3)$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8(x + 3) = a(x + 3)$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of a is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{a(x+3)}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of x . If $x = 0$, then substituting 0 for x in the given equation yields $4(0) + 12 = \frac{a(0+3)}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of a is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of a is 3, the given equation is equivalent to $4x + 12 = \frac{3(x+3)}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{9}{2}$, which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of a is 12, the given equation is equivalent to $4x + 12 = \frac{12(x+3)}{2}$, or $4x + 12 = 6x + 18$, which has one solution, not infinitely many solutions.